

VIRTUAL QUAKE

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THE PROBLEM

 Mexico City, Mexico

1985: Magnitude 8.0 Earthquake Leaves 250,000 People Homeless

2017: Magnitude 7.0 Earthquake

12 Days Later and 75 miles Away Puebla , Mexico
Magnitude 8.2 Earthquake

THE AFTERMATH

MEXICO CITY IS NOT ALONE

Mega-cities worldwide are home to millions of people, and tend to be high earthquake risks.

01

184,000 Homes Destroyed
370 Lives Lost

02

\$150 million dollars granted to
rebuild homes

03

No new preventative measures if
something happens again.

OUR SOLUTION

THE FIRST VIRTUAL EARTHQUAKE SIMULATION
APPLICATION TO BE USED PRE-CONSTRUCTION TO
MITIGATION FUTURE DAMAGE

- 1) Earthquake Simulation Using Seismic Data
- 2) AI-Driven Optimization



01 User fills out required parameters.



02 User runs the simulation.



03 AI generates optimization recommendations during simulation events.



04 Contextualized damage report that applies AI suggestions with cost effectiveness.

HOW IT WORKS

WE WON'T BORE YOU WITH THE MATH

$$[C] = 2\xi\omega_n[M]$$

where ξ is the damping ratio and $\omega_n = \sqrt{k_{story}/m}$ is the natural frequency.

$$\begin{aligned} [K_{eff}] &= [K] + \frac{\gamma}{\beta\Delta t}[C] + \frac{1}{\beta\Delta t^2}[M] \\ \{p_{eff}\} &= -[M]\{1\}\ddot{u}_g(t_i) + [M]\left(\frac{1}{\beta\Delta t^2}\{u_{i-1}\} + \frac{1}{\beta\Delta t}\{\dot{u}_{i-1}\} + \left(\frac{1}{2\beta} - 1\right)\{\ddot{u}_{i-1}\}\right) \\ &+ [C]\left(\frac{\gamma}{\beta\Delta t}\{u_{i-1}\} + \left(\frac{\gamma}{\beta} - 1\right)\{\dot{u}_{i-1}\} + \Delta t\left(\frac{\gamma}{2\beta} - 1\right)\{\ddot{u}_{i-1}\}\right) \\ \{u_i\} &= [K_{eff}]^{-1}\{p_{eff}\} \end{aligned}$$

$$\{\ddot{u}_i\} = \frac{1}{\beta\Delta t^2}(\{u_i\} - \{u_{i-1}\}) - \frac{1}{\beta\Delta t}\{\dot{u}_{i-1}\} - \left(\frac{1}{2\beta} - 1\right)\{\ddot{u}_{i-1}\}$$

$$\{\dot{u}_i\} = \{\dot{u}_{i-1}\} + \Delta t[(1 - \gamma)\{\ddot{u}_{i-1}\} + \gamma\{\ddot{u}_i\}]$$

where $\beta = 0.25$ and $\gamma = 0.5$ correspond to the average-acceleration Newmark method.

$$IDR_i = \frac{|u_i - u_{i-1}|}{h_i} \quad (4)$$

Inter-story Drift Ratio (IDR) measures how much a building bends during the earthquake by calculating how much one floor sways relative to the floor below it, divided by the height between the floors.

$$a_{peak,i} = \max_t |\ddot{u}_i(t)| \quad (5)$$

Peak Floor Acceleration (*a_{peak}*) is the greatest amount of acceleration a floor in the building experiences during the earthquake.

$$[M]\{\ddot{u}(t)\} + [C]\{\dot{u}(t)\} + [K]\{u(t)\} = -[M]\{1\}\ddot{u}_g(t) \quad (1)$$

where $[M]$, $[C]$, and $[K]$ are the mass, damping, and stiffness matrices; $\{u(t)\}$, $\{\dot{u}(t)\}$, and $\{\ddot{u}(t)\}$ are the displacement, velocity, and acceleration vectors of ⁽³⁾ floors; and $\ddot{u}_g(t)$ is the ground acceleration input.

$$D = \min\left(1, \frac{\max(IDR_i)}{0.02}\right) \quad (6)$$

WHAT MAKES US UNIQUE

FIRST APPLICATION TO IMPLEMENT THIS

No other application does what we do.

COMPUTATIONAL EFFICIENCY

Lump-mass model keeps it light
compared to similar applications.

PRE-CONSTRUCTION IMPLEMENTATION

Used before a building is built to
reduce risk.

ACCESSIBILITY AND TRANSPARENCY

AI performance balances
optimization for safety with cost
effectiveness to support inclusive
city planning, and walks you
through every decision.

Bonus: it's explainable!

SCALABILITY AND FEASIBILITY

Geographically adaptable.

Computational efficiency makes application accessible even on modest hardware.

Particularly important for cities with less money in their budget.

Can be scaled to include more input parameters, as determined by the user.

- Such as different building types.

Parameters can be easily found in building designs and structural codes.

Running simulations in parallel for large-scale projects in infrastructure development

AI model expansion on optimization decisions.



DEMO TIME :)